NETFLOW

ECONOMY MODEL

LIQUID FLOW MEASUREMENT · NT-VMF-EKO

STANDARD SERIES ELECTROMAGNETIC FLOW METER

A simple, reliable and cost-effective solution for basic measurement and automation needs



0,1 – 15 m/s	±%0,3 R	220 VAC / 24 VDC	MODBUS RTU
VELOCITY · RESOLUTION 1 mm/s	BEST ACCURACY	BESLEME · ±%10	STANDART · HART OPS.

The NT-VMF-EKO measures the flow rate of conductive liquids with its low-frequency square-wave excitation converter. It does not require additional electrodes or grounding rings for empty pipe detection. In addition to analog, frequency, pulse and alarm outputs, Modbus communication is provided as standard; it can be used with a compact or remote (wall-mount) converter.

FUNCTIONS

OUTPUTS

0 – 10 mA veya 4 – 20 mA analog
1 – 5000 Hz frekans
Adjustable pulse
Photoelectric isolation >1000 VDC

ALARMLAR

Empty pipe · excitation (system)
High and low limit
Adjustable by flow, percent, forward, reverse or net flow

TOPLAM SAYICILAR

Forward totalizer
Reverse totalizer
Net totalizer (forward – reverse)
Password-protected reset

COMMUNICATION

Modbus RTU, RS-485 / RS-232
1000 V izolasyon
HART (optional)

LINEARIZATION

4-point non-linear correction at low velocities

DATA SECURITY

Factory data backup and restore
Parameter and totalizer transfer to external EEPROM

MEASUREMENT ACCURACY

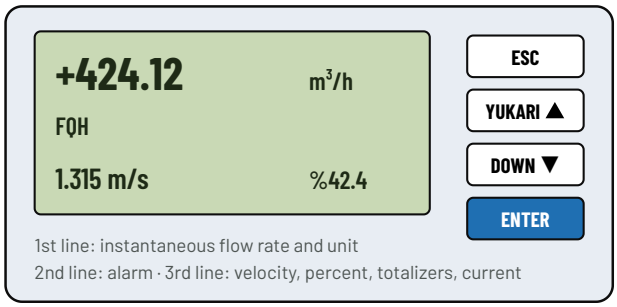
DIAMETER	< 0,3 m/s	0,3 – 1 m/s	1 – 15 m/s
DN3 – DN20	±%0,25 TS	±%1,0 R	±%0,5 R
DN25 – DN600	±%0,25 TS	±%0,5 R	**±%0,3 R**
DN700 – DN3000	±%0,25 TS	±%1,0 R	±%0,5 R

TS: full scale · R: reading

TECHNICAL SPECIFICATIONS

Excitation	Low-frequency square wave · 5 / 4,167 / 3,125 Hz
Excitation current	125 mA
Flow velocity	0,1 – 15 m/s · resolution 1 mm/s
Power supply	220 VAC ±%10 · 24 VDC ±%10
Power consumption	< 10 W (sensor included)
Analog output	0 – 10 mA (load 0 – 1200 Ω) · 4 – 20 mA (load 0 – 600 Ω) · ±0.1% ±10 µA
Frequency output	1 – 5000 Hz
Pulse output	0,001 – 1,000 m^3 veya L / puls · maks. 50 ms
Digital output load	Harici besleme maks. 24 VDC · 100 mA
Communication	Modbus RTU (RS-485 / RS-232) · HART ops.
Display	4-line LCD · 4 keys · Turkish / English
Operating temperature	–20 ... +70 °C
Relative humidity	%5 – %90
Mains frequency	50 / 60 Hz selectable

DISPLAY AND KEYS

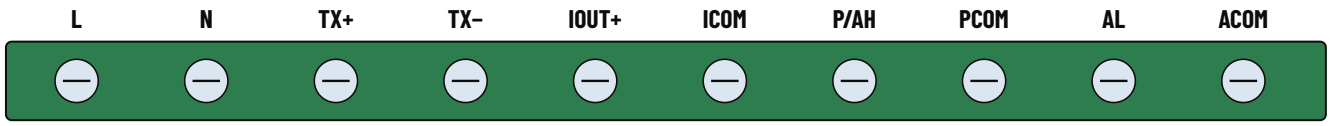


1st line: instantaneous flow rate and unit
2nd line: alarm · 3rd line: velocity, percent, totalizers, current

FQH	Flow high limit alarm
FQL	Flow low limit alarm
FGP	Empty pipe alarm
SYS	System (excitation) alarm

Press ENTER key for 3 seconds to open the password screen. If no key is pressed for 10 minutes, the device returns to the measurement screen.

CONNECTION TERMINALS KOMPAKT CONVERTER



Power supply | RS-485 A / B | Current output | Frequency · pulse · high alarm | Low alarm · flow direction

L / N: 220 VAC phase / neutral or 24 VDC + / –. In remote type converter, additionally sensor excitation (CD1, CD2) and signal (SIG1, SIG2, SGND, DS1, DS2) terminals are present.

TROUBLESHOOTING

SYMPTOM	KONTROL
No display image	Check the power supply connection, fuse and supply voltage.
Empty pipe alarm	Check signal cables and that the pipe is completely full. Clean the electrodes if dirty; if conductivity is low, increase the empty pipe coefficient.
Excitation alarm	Check excitation and electrode cables. If the problem persists, contact technical service.
Unstable measurement	Check grounding, signal cables and sensor coefficients.

OUTPUT SETTINGS

KLEMENS	MOD	BEHAVIOR
P/AH · PCOM	**Frekans**	0 – 5000 Hz, proportional to flow scale. Recommended for control applications.
	Puls	Each pulse corresponds to the set quantity. Recommended for totalizing and batching.
	High alarm	Normally open or normally closed selectable.
AL · ACOM	**Flow direction**	Low: negative direction · High: positive direction
	Low alarm	Normally open or normally closed selectable.
IOUT+ · ICOM	**Current**	0 – 10 mA or 4 – 20 mA, internal 24 VDC, max. 600 Ω at 4 – 20 mA. 4 mA and 20 mA points are fine-adjustable from the menu.

FREKANS

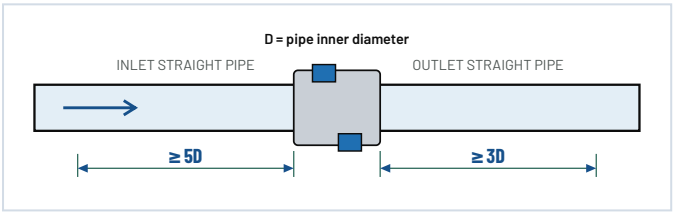
$$F = (Q / Q\ \text{full scale}) \times F\ \text{skala}$$

VOLUMETRIC FLOW

$$Q\ (m^3/s) = 0{,}0007854 \times D^2 \times v \times 10^{-3}$$

D: internal diameter (mm) · v: flow velocity (m/s). Pulse coefficient must be selected so that output frequency does not exceed 5000 Hz. Digital outputs drive 16 mA with 1.5 kΩ resistance; external load max. 100 mA.

INSTALLATION KURALLARI



- The measuring tube must always be completely full.
- In vertical lines, the flow must be from bottom to top.
- Install on the discharge side of the pump and inlet side of the control valve.
- Position so that the electrode axis is horizontal.
- Grounding: to PE terminal with copper conductor, cross-section > 1.6 mm², resistance < 10 Ω.
- For remote type, use grey dual-shielded cable for the signal line; the shield end is connected to SGND.

TÜRKAK
ISO/IEC 17025:2017

ACCREDITATION	UNCERTAINTY	SCOPE	RECOGNITION
AB-0354-K	**±%0,27**	**1 – 350 m³/h**	**EA · ILAC MRA**

Every flow meter is calibrated in our TÜRKAK accredited calibration laboratory before shipment; a traceable calibration certificate is shipped with the device.

Contact our technical team for diameter, connection and output selection.

+90 541 373 79 53
info@netflw.com